

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.1
```

```
## Warning: package 'tidyr' was built under R version 4.4.1
```

```
library(lubridate)
library(here)
```

```
## Warning: package 'here' was built under R version 4.4.1
```

```
Neonics <- read.csv(file = here('./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = T)
Litter <- read.csv(file = here('./Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = T)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: My guess is that neonicotinoid may include some chemical components that are difficult to be degraded, and thus remain in earth, which gradually affect biotas through bio-accumulative effects. Indeed, neonicotinoids kill pest by manipulating their nerves, unfortunately, they kill beneficial pollinators at the same time. They can remain in soil for years, bringing threats to human, agriculture and ecosystem.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: By collecting and analyzing litter and woody debris, researchers may examine tree species exist in a certain area, and study the chemistry and isotope of them.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. The location of sampling site is Tower plot, they set 20 plots in forested area, while 30 plots in less-vegetation-covered area, then randomly select at least 2 subplots within plots. 2. One elevated trap and one ground trap is deployed for every 400 m² plot area, resulting in 1-4 trap pairs per plot, and can be placed targetted or randomized. 3. Plot edges must be separated by a distance 150% of one edge of the plot, as well as away from large paved roads, infrastructure, streams etc.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics) # 4623 30
```

```
## [1] 4623 30
```

```
dim(Litter) # 188 19
```

```
## [1] 188 19
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
sort(summary(Neonics$Effect))
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)
##           1           5           7           9
## Biochemistry Accumulation Intoxication Immunological
##          11          12          12          16
## Morphology      Growth      Enzyme(s)      Genetics
##          22          38          62          82
## Avoidance      Development Reproduction Feeding behavior
##         102         136          197         255
## Behavior      Mortality      Population
##         360         1493         1803
```

Answer: The most common one is Population with 1803 records, follows by Mortality with 1493 records. Because these are effects that people concern the most according to published articles collected.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##         667         285         183
## Carniolan Honey Bee      Bumble Bee      Italian Honeybee
##         152         140         113
##      (Other)
##        3083
```

Answer: They are 1) Honey Bee, 2) Parasitic Wasp, 3) Tailed Bumblebee, 4) Carniolan Honey Bee, 5) Bumble Bee and 6) Italian Honeybee. They are the species of the most concern and effected by neonics, knowing the species may help with further research and conservation work.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

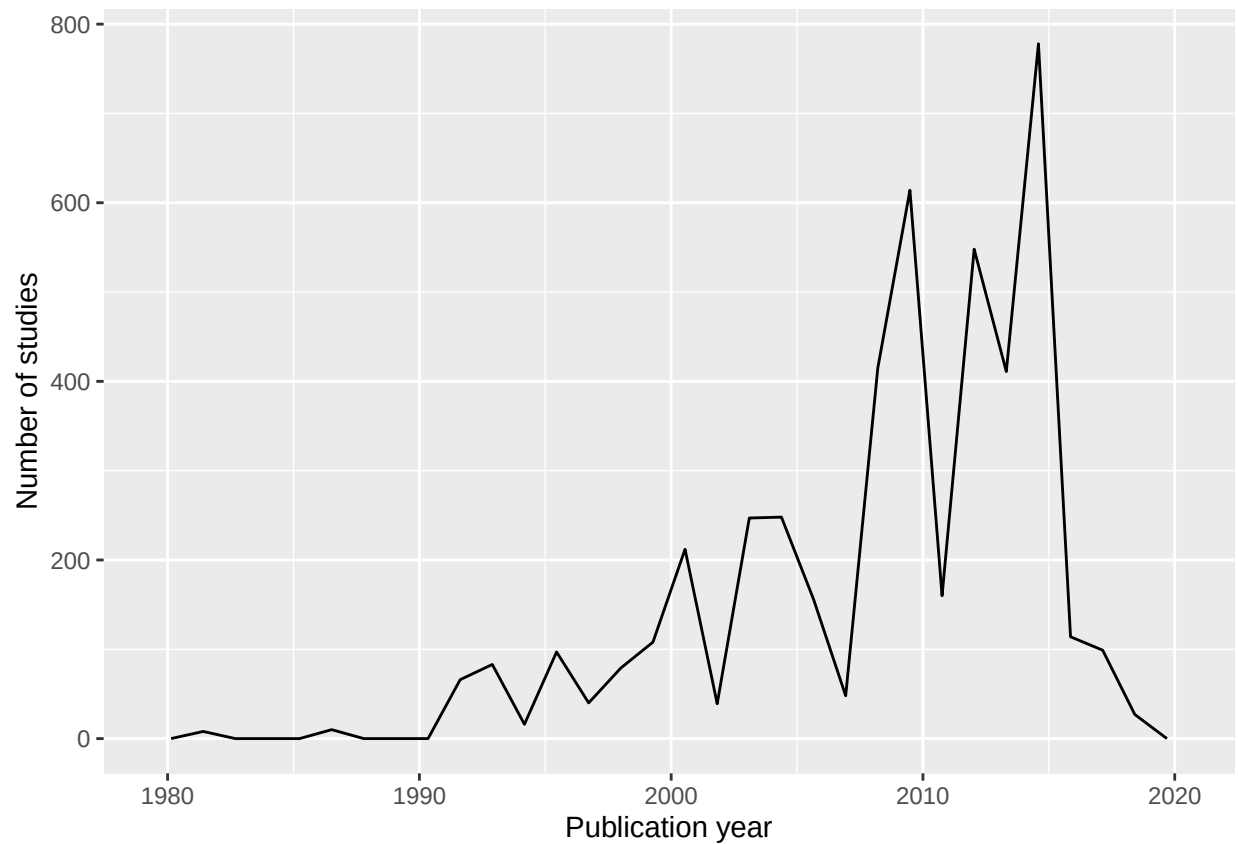
Answer: It is factor. Despite most values in this column look numeric, but there are also values like “NR”, “~10”, “12/” etc., which cannot be stored as numeric values.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics)+  
  geom_freqpoly(aes(Publication.Year))+  
  ylab("Number of studies")+  
  xlab("Publication year")
```

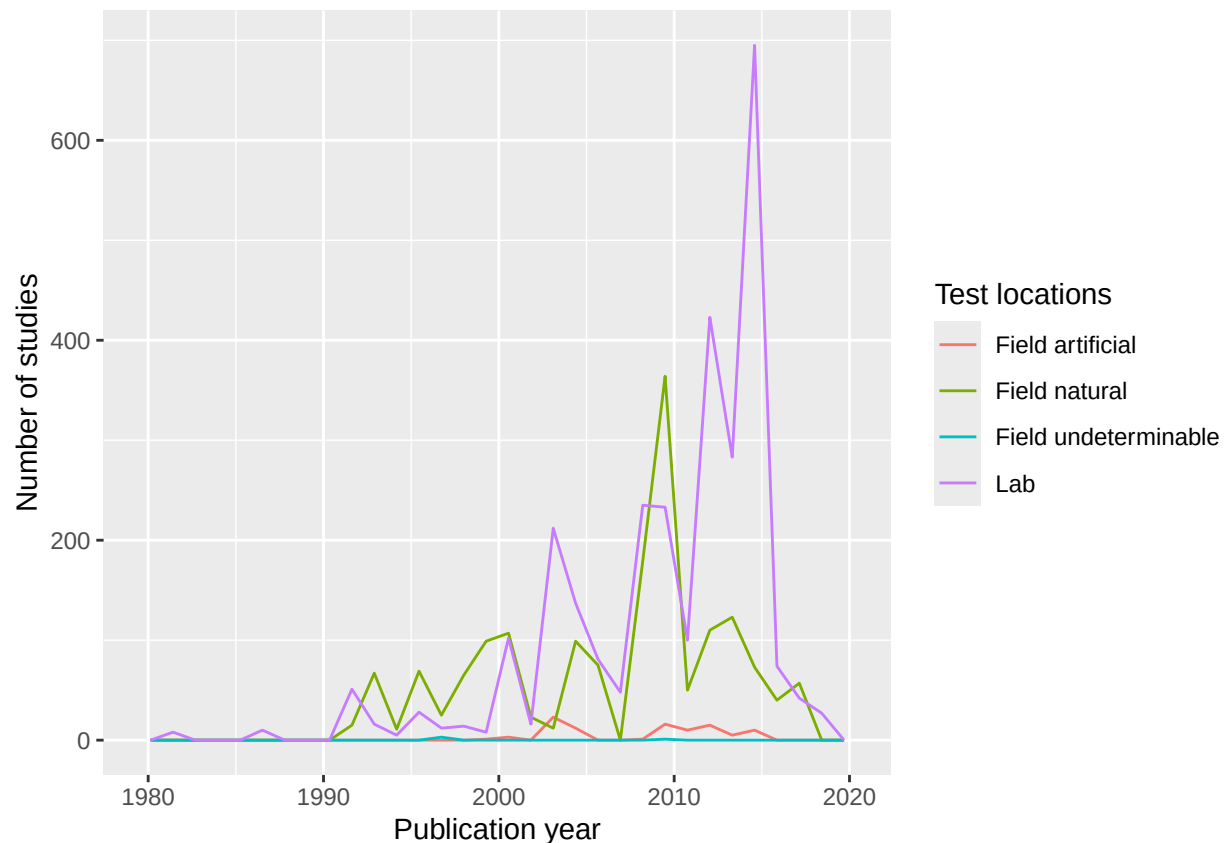
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics)+  
  geom_freqpoly(aes(Publication.Year, colour = Test.Location))+  
  ylab("Number of studies")+  
  xlab("Publication year")+  
  labs(colour = "Test locations")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



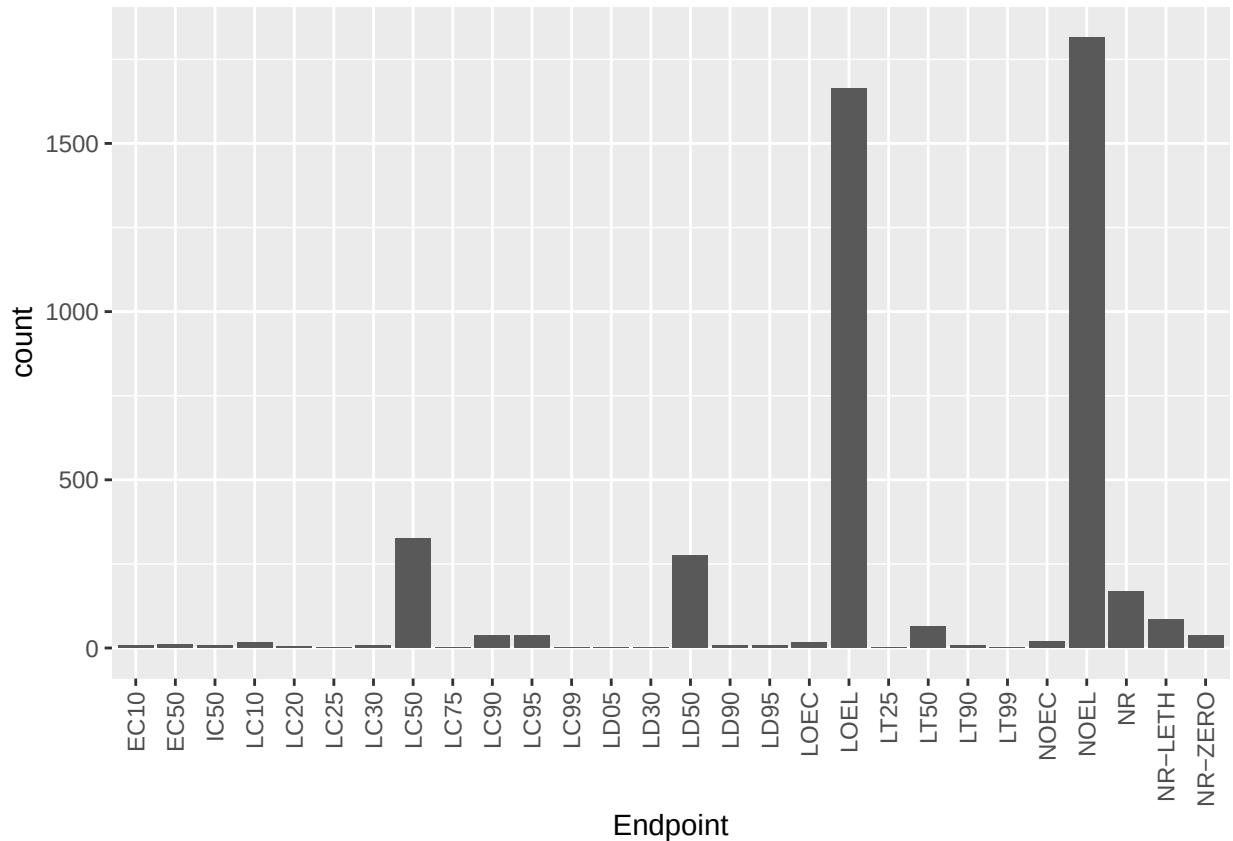
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: Lab studies rose significantly after 2010 and dominated around 2015, reaching over 600 studies. Field natural studies also rose after 2000, peaking around 2010 with around 300–350 studies.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics)+
  geom_bar(aes(x=Endpoint))+
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: LOEL and NOEL. LOEL stands for “Lowest-observable-effect-level” — lowest dose (concentration) producing effects that were significantly different (as reported by authors) from responses of controls (LOEAL/LOEC) NOEL stands for “No-observable-effect-level” — highest dose (concentration) producing effects not significantly different from responses of controls according to author’s reported statistical test (NOEAL/NOEC)

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate) #factor
```

```
## [1] "factor"
```

```
Litter$collectDate <- as.Date(Litter$collectDate, format = "%m/%d/%y")
unique(Litter$collectDate) #"2018-08-02" "2018-08-30"
```

```
## [1] NA
```

- Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

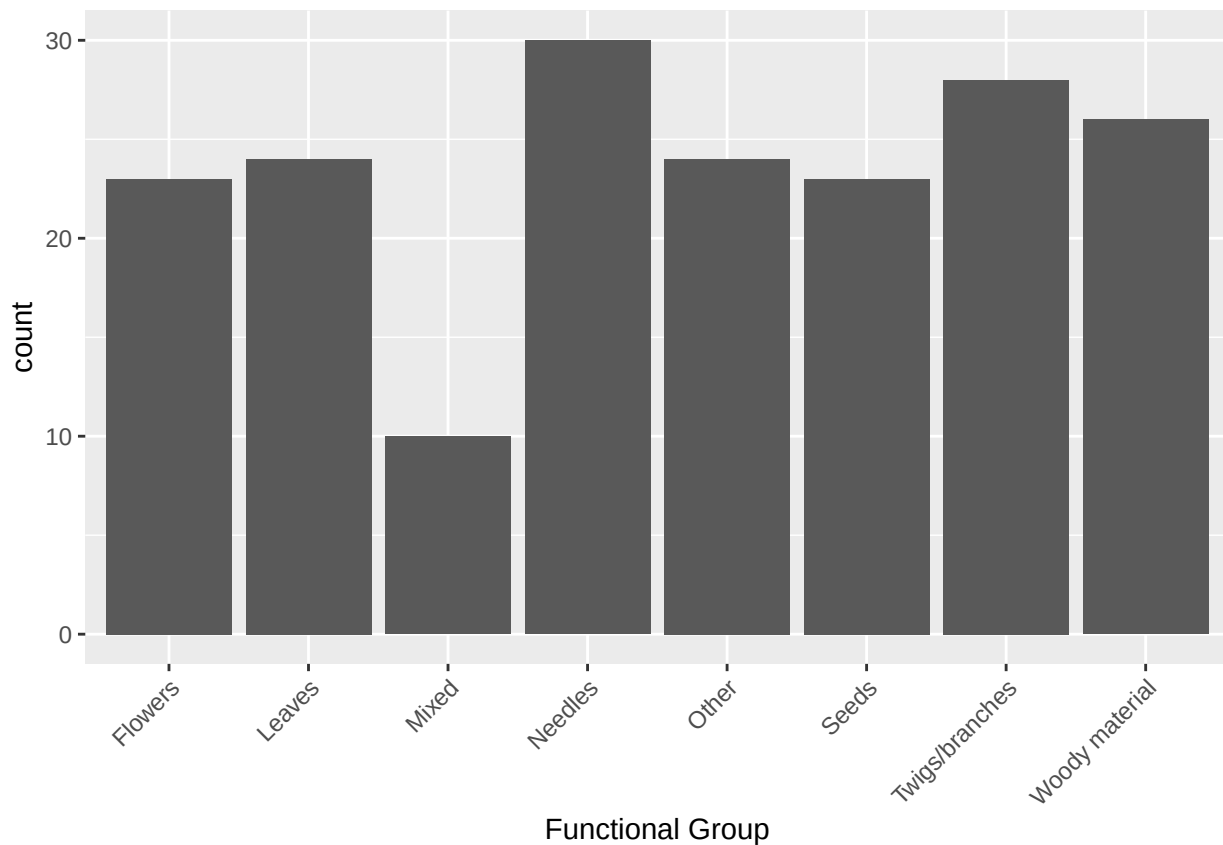
```
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14       8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

Answer: Unique() returns a vector which lists the factors in the plotIDs column, with duplicate elements removed; while summary() summarize the information, providing names and counts of elements in this column.

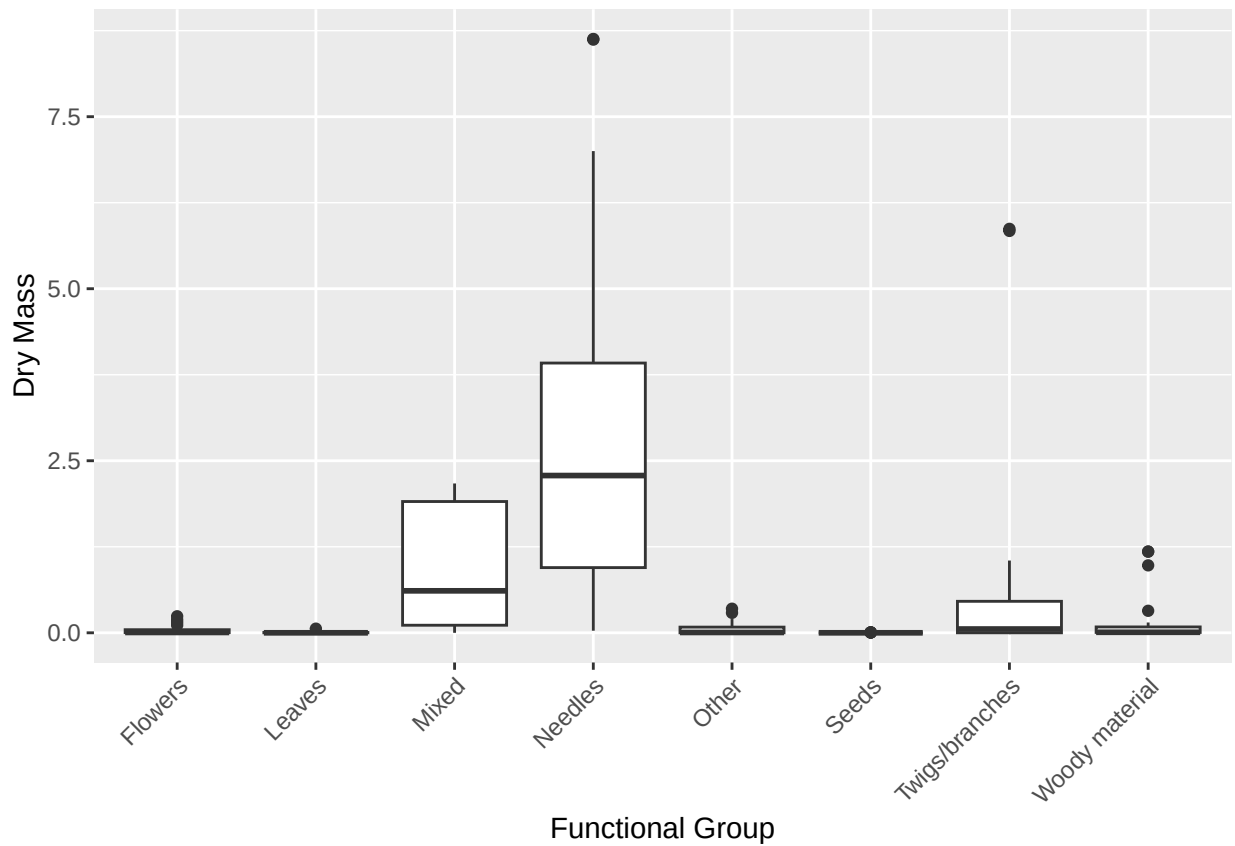
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(Litter)+  
  geom_bar(aes(functionalGroup))+  
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+  
  xlab("Functional Group")
```

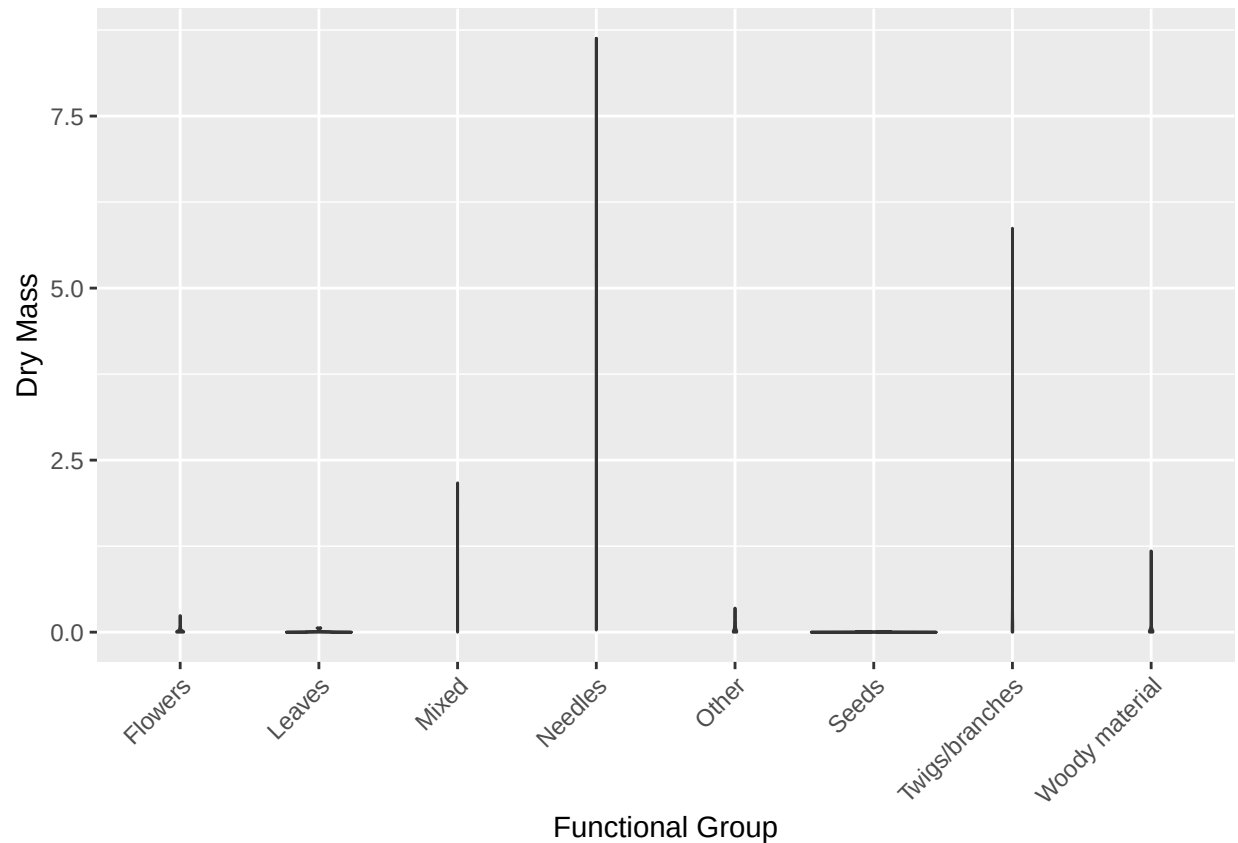


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by functional-Group.

```
ggplot(Litter)+  
  geom_boxplot(aes(y = dryMass, x = functionalGroup))+  
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+  
  xlab("Functional Group")+  
  ylab("Dry Mass")
```



```
ggplot(Litter)+  
  geom_violin(aes(y = dryMass, x = functionalGroup))+  
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+  
  xlab("Functional Group")+  
  ylab("Dry Mass")
```

Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: Because in the dryMass column, most values differ from each other, so a distribution of values may not be obvious for a violin plot, thus ending up in a single string in a violin plot. But box plots provide the overall range and quartiles of the values, which is more effectively visualized here.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles have the highest biomass, with an average around 2.5.